OM GOVIND JHA

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Research Interests

Multimodal Machine Learning & Affective Computing, Continuous Flow Matching & Generative Modeling, Clustered Dynamic Sparse Federated Learning, Interpretable AI (SHAP), Convex Optimization & Game-Theoretic Statistical Inference.

Education

Indian Institute of Science (IISc) — Bengaluru, India
M.Tech. in Robotics and Autonomous Systems — *2026 – 2028*
- **Focus Areas:** Autonomous Robotics, Robot Perception, Deep Learning, State Estimation, Optimal Control.

Indian Institute of Science Education and Research (IISER) — Bhopal, India
B.S. in Computer Science and Engineering, Minor in Data Science; ***CPI: 8.72 / 10.0*** — *2022 – 2026*
- **Core Coursework:** Federated Learning, Deep Learning, Computer Vision, Robotic Perception, Machine Learning, NLP, Applied Optimization, Probability & Statistics, Linear Algebra, Real Analysis, DBMS, Data Structures & Algorithms.

Chinmaya Vidyalaya (Class XII, CBSE) — Bokaro Steel City, India
All India Senior School Certificate Examination (AISSCE); ***Score: 94.2%*** — *2022*

DAV Public School (Class X, CBSE) — Jhanjharpur, Bihar, India
All India Secondary School Examination (AISSE); ***Score: 96.4%*** — *2020*

Publications & Preprints

- *Author Contribution Note:* ***Om Govind Jha is the primary / first author on all 4 publications listed below.***

Peer-Reviewed Conference Proceedings

Modeling Affective Distress from Video: A Modality-Aware Multi-Branch Learning Approach — Napoli, Italy
Om Govind Jha, Nilesh Kumar Sahu, Haroon R. Lone.
*Proceedings of the 28th ACM International Conference on Multimodal Interaction (**ICMI '26**)*, October 2026.
DOI: 10.1145/3776574.3831196 | Direct PDF: *ACM DL Link* | Code: *github.com/OmGovind15/MAMBL.*

Accepted to Appear

Teacher-Guided Discriminative Flow Matching for Latent Representation Learning — Tromsø, Norway
Om Govind Jha, Manoj Bamniya, Ayon Borthakur.
*Proceedings of the Northern Lights Deep Learning Conference (**NLDL 2027**).* **Status: Accepted / To Appear**.
Camera-ready version / preprint available upon request (proceedings to appear).

Journal Articles Under Review

Predictive Modeling in ODI Cricket: Training Strategy, Feature Engineering, & Explainability — Under Review
Om Govind Jha, Kundan Kandhway.
International Journal of Sports Science & Coaching (**SAGE Publications**). **Status: Revision Stage (1st Round Completed)**.
Preprint / Working Paper available upon request.

Conference Preprints & Submissions

CA-AFP: Cluster-Aware Adaptive Federated Pruning for Heterogeneous Federated Learning — Under Review
Om Govind Jha, Harshit Shukla, Haroon Rashid Lone.
Submitted to *ACM IKDD Conference on Data Science and Management of Data (**CoDS-COMAD 2026**)*.
Submitted manuscript / preprint available upon request.

Research Appointments

Research Intern | *Supervisor: Dr. Ayon Borthakur* May 2025 – Mar 2026
Indian Institute of Technology (IIT) Guwahati *Guwahati, India*
- Developed a teacher-guided Discriminative Flow Matching (DFM) framework by parameterizing independent conditional vector-field predictors on shared deep backbones.
- Formulated classification as a continuous probability transport path pairing strongly augmented student embeddings with EMA teacher targets; evaluated on CIFAR-100 (top-1: 69.38%) and Tiny-ImageNet (top-1: 63.50%).
- Engineered local block-wise training combining step-wise label denoising with local loss propagation for Vision Transformers (ViT) and DETR, mitigating backpropagation memory overhead.

Research Intern | *Supervisor: Dr. Kundan Kandhway* Nov 2024 – Mar 2025
Indian Institute of Science Education and Research (IISER) Bhopal *Bhopal, India*
- Constructed an end-to-end forecasting pipeline analyzing 1,394 international ODIs (2002–2024); engineered 11 dynamic innings-specific metrics (Pressure Index, Target Deficit, Run-Rate Differential).
- Formulated a chronological expanding-window scheme that completely prevents temporal data leakage across sequential matches.
- Demonstrated that a Unified Inning Model excels in 1st innings score modeling, whereas an adaptive Over-Wise Model Switching strategy achieves superior 84.6% accuracy in 2nd innings; evaluated game-theoretic feature importance using SHAP.

Summer Research Fellow | *Supervisor: Dr. Rajesh Sundaresan* May 2024 – July 2024
Indian Institute of Science (IISc), Bengaluru *Bengaluru, India*
- Recipient of prestigious INSA-IASc-NASI fellowship; researched optimal survey design methodologies for population disease burden estimation.
- Formulated convex optimization programs utilizing Fisher Information Matrices and Maximum Likelihood Estimation (MLE) under operational testing budget constraints.
- Analyzed zero-sum game-theoretic minimax formulations for robust worst-case prevalence bounds across stratified demographic groups.

Selected Research Projects

MAMBL: Modality-Aware Multi-Branch Learning for Video Distress Detection | *PyTorch, MediaPipe* Dec 2025 – May 2026
- Engineered an automated diagnostic pipeline separating heterogeneous temporal behavioral cues: routed sparse Action Units to 1D CNNs and smooth pose/gaze dynamics to BiLSTMs based on sparsity and autocorrelation diagnostics.
- Formulated gated temporal attention pooling and gradient-based spatial saliency, achieving 71.04% accuracy on a clinical Trier Social Stress Test dataset and competitive performance across StressID, UBFC-Phys, and AVEC 2017.

CA-AFP: Cluster-Aware Adaptive Federated Pruning for Heterogeneous FL | *PyTorch, Flower* Aug 2025 – Feb 2026
- Proposed a clustered dynamic sparse federated learning framework allowing heterogeneous client groups to independently evolve sparse network topologies under strict communication and memory budgets.
- Clustered clients via optimization trajectory cosine similarity; implemented adaptive magnitude pruning and gradient-guided regrowth achieving Pareto-optimal accuracy under a 70% parameter budget on CIFAR-10, FEMNIST, and WISDM.

Edge Vision: Vehicle Detection and Perspective Speed Tracking | *YOLOv8, OpenCV, Python* Sep 2024 – Nov 2024
- Implemented an automated video analytics system applying YOLOv8 vehicle detection with 2D-to-3D perspective projection calibration for metric speed estimation and automated traffic violation alerts.

Honors, Fellowships & Distinctions

Reliance Foundation Undergraduate Scholar (2022 – 2026): Awarded prestigious national merit scholarship granted to top engineering undergraduates across India for academic excellence and leadership.

INSA-IASc-NASI Summer Research Fellowship (2024): Competitive national fellowship awarded by the Indian National Science Academy to pursue funded research at IISc Bengaluru.

GATE Data Science & AI (DA) Qualified (2025): Qualified national engineering graduate aptitude examination with high percentile.

JEE Advanced (2022) & IMU-CET (2022): Ranked in top 1% among 1.2+ million applicants in JEE Advanced; secured AIR 98 in IMU-CET.

Teaching, Service & Academic Leadership

Teaching Assistant, *Data Science in Practice* (Jan 2025 – May 2025), Department of EECS, IISER Bhopal: Mentored 60+ undergraduate students, conducted weekly laboratory problem sessions, designed programming assignments in Python and Scikit-learn, and held regular office hours.

Student Coordinator, Economics Club, IISER Bhopal (2024 – 2025): Led student workshops on quantitative modeling, game theory, and algorithmic decision analysis.

Technical & Computational Skills

Programming Languages: Python, C, C++, SQL, Bash, MIPS Assembly, LaTeX
Machine Learning & Deep Learning: PyTorch, Torchvision, Flower, Hugging Face, OpenCV, YOLOv8, Scikit-learn, NumPy, Pandas, SciPy, SHAP
Core Theoretical Competencies: Continuous Flow Matching, Diffusion Models, Clustered Federated Learning, Vision Transformers, Optimal Transport, Convex Optimization, Statistical Inference (MLE, Hypothesis Testing), Game Theory
Developer Tools & Platforms: Git, Linux/Unix, Docker, Google Colab, Jupyter, Weights & Biases, AutoCAD, MATLAB